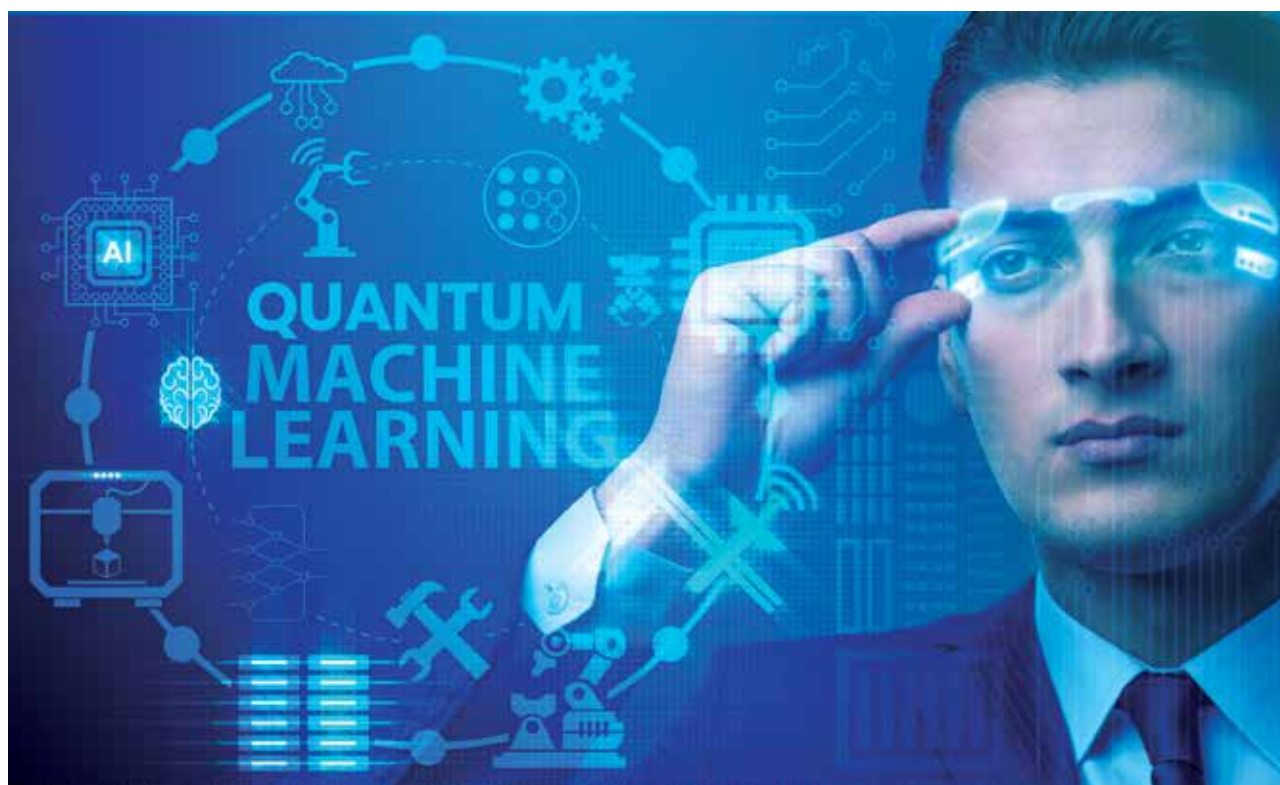


Quantum Machine Learning: Writing Quantum Algorithms for AI Models

Quantum machine learning combines the strengths of quantum computing and artificial intelligence to solve problems faster and more efficiently than traditional computing methods. By leveraging quantum mechanics, QML has the potential to revolutionise areas such as data analysis, image processing, and natural language understanding. Let's understand the basics of QML and learn how to write quantum algorithms for AI models.



Quantum computing operates on quantum bits or qubits. Unlike classical bits, which are either 0 or 1, qubits can exist in a superposition of states. This allows quantum computers to process multiple possibilities simultaneously, making them exponentially faster for certain tasks.

Machine learning uses algorithms to learn patterns from data. Quantum machine learning (QML) enhances these algorithms by utilising quantum computing principles, such as:

- **Superposition:** Processing multiple inputs at once.
 - **Entanglement:** Strong correlations between qubits to improve accuracy.
 - **Quantum interference:** Enhancing desired outputs while suppressing errors.
- Key quantum algorithms used in AI include:
- **Quantum data encoding:** Converts classical data into a quantum format for processing.
 - **Quantum support vector machines (QSVM):** Enhances classification tasks.
 - **Quantum neural networks (QNN):** Accelerates the training of neural networks.
 - **Variational quantum circuits (VQC):** Solves optimisation problems in machine learning.